

Sign Language Recognition using Machine Learning Algorithm

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Abstract—Sign Language Recognition System aims to rephrase sign language into text or speech to ease communication between deaf and hearing individuals. This issue has far-reaching implications but remains exceptionally hard due to the complexity and large variation of hand movements. Existing SLR styles use palm-drafted characteristics to determine the shift of sign language and make classification models based on these features. Still, it's delicate to design dependable features that accommodate wide variations in gestures. KNN (k-nearest neighbor) algorithm has shown major challenges like curse of dimensionalities, large dataset sizes, class imbalance, noise, and outliers. To address this conclusion, this study suggests a new convolutional neural network (CNN) that can automatically extract discriminative spatial-temporal elements extracted without information known from unprocessed video streams and avoid designing features. To improve performance, multi-channel video streams (such as color data, depth cues, and body joint orientations) are sent into CNN as input to combine color, depth, and movement data. Using a real-world dataset acquired using Microsoft Kinect, this study evaluated the proposed model and showed its effectiveness over traditional methods based on manual labor feature.

Keywords— *convolutional neural network, sign-recognition, hand movements*

I. INTRODUCTION

Sign language is one of the most commonly used communication methods for hearing-impaired people, expressed through changes in hand shapes, body movements and even facial expressions. Sign language recognition remains a very challenging task due to the difficulty in sharing information from hand shapes and body trajectories [1][2]. This study postulates a useful recognition model to aid deafness individuals in communicating with hearing individuals by translating sign language into written and speech. Practically, the biggest difficulty in sign language development of characteristics to represent the shape and trajectory of the hand. Hand form description, in particular, entails tracking hand areas in video streams and segmenting hand shape imagery from composite images. Background and gesture recognition issues for each frame. Trajectories are also concerned with key point tracking and curve matching. Although a lot of research has been done on these

two topics so far, it is still difficult to achieve satisfactory results for Sign Language Recognition due to the fluctuation and interference of hand and body joints. Furthermore, integrating hand features and network features is a non-trivial problem [2]. To address these adversities, this paper proposing CNNs to naturally incorporate hand forms, trajectories, and facial looks. instead than utilising the standard colour image as the inputs of the network, use the color image, deep image and hand skeleton image as input at the same time all powered by Microsoft Kinect. It is a motion sensor. Using the public Windows SDK, the positions of body joints can be retrieved in real time. Therefore, select Kinect as the acquisition medium for recording the character word dataset. Changing color and depth at the pixel level is useful information to differentiate between different drawing operations. These changes in the time dimension of the body joints can represent the trajectory of the character's movements. Utilising a variety of picture resources as input causes the CNN to focus not only on color changes but also on depth and trajectory. Notably, since CNNs are able to automatically learn features from raw data without prior knowledge, by-passing the challenges of monitoring hands, separating hands from the background, and developing hand descriptions [3]. In recent years, CNNs have been applied to the classification of video streams. A potential problem with CNNs is that it is time consuming. Training a CNN with millions of videos takes weeks or months. Fortunately, it is still possible to achieve real-time efficiency. This study proposes to derive both temporal and spatial details from web cams for sign language recognition (SLR) using Convolutional Neural Network. Existing SLR techniques use hand- drafted features to describe the stir of sign language and produce classification models according to these features. CNNs, however, can automatically record motion data from raw video data. Avoid design features. This study develops a CNN that takes variety of data as inputs. Through convolution and subsampling neighbouring video frames, the architecture incorporates colors, depth, and trajectory information. Experimental results show that 3D CNN can significantly outperform Gaussian Mixture Model with Hidden Markov Model (GMM-HMM) baseline for some character words that were recorded [4].

II. LITERATURE SURVEY

Zaid Omar et al. [5] stated that the steps of gesture recognition include data collection, pre-processing, segmentation, feature extraction, and classifications. Whereas dynamic sign languages use video, which is a continuous stream of frames of pictures, static gesture recognition uses single frames of images as its input. Vision-based methods differ from sensor-based methods primarily in the way that data is acquired. This section focuses on the methods and procedures utilised by studies on vision-based gesture recognition.

Rafiqul Zaman Khan et al. [6] used non-geometric characteristics and a multivariate Gaussian distribution to identify hand movements. Two strategies are used to segment the input hand image: Skin tone-based segmentation using the HSV colour model and histogram noise removal techniques. Some operations are carried out in the interest of acquiring palm features, capture the hand's shape; The modified Direction Analysis Algorithm is used to compute object (hand) slope and trend by identifying the orientation of the wrist movement. It is utilised in order to determine a linkage between statistical parameters from the data.

According to Sukanya Dessai et.al. [7] A machine's ability to comprehend people behaviour and its significance can be used in a variety of situations. The study of sign language recognition is one topic that has aroused interest. Those who are deaf or dumb use sign language to communicate. They are unable to communicate with the outside world without the aid of sign language interpreters. This study reviews the methods used in current research on sign language recognition. The approaches are studied at the phases of gathering data, pre-processing, segmenting, obtaining features, and classifying it. The aforementioned literature review is conducted individually at each stage. Some of the most popular classification methods used for recognition include ANN, SVM and fuzzy inference systems.

III. METHODOLOGY

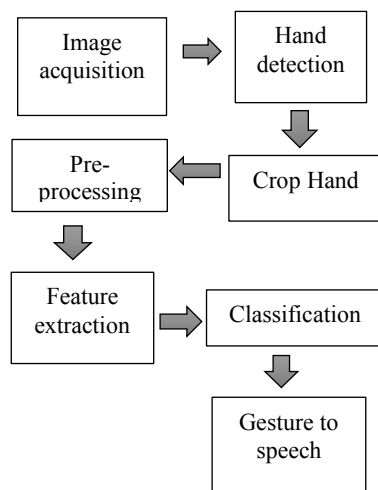


Fig.1. Flow chart of the suggested work

A. Image Acquisition

It's the initial and most important phase of the gesture recognition process. webcam connection is required in order

to capture images using a Webcam. Several laptops now include an inbuilt camera system, which aids in the capture of photos for further processing. The embedded camera can identify hand gesture and position by capturing gestures.

B. Image Pre-processing

This phase involves retouching, filtration, adjusting intensity and tone and many other operations are included in image pre-processing. Picture enhancing, picture clipping, and picture thresholding are employed in this procedure. Images that are captured are in RGB format. Hence, the primary task is to transform RGB photos to binary information, and then crop the image to get rid of the undesired parts. [8][9].

C. Feature Extractions

This is an essential process in building the dataset for hand gesture interpretation. Both the global and local visual characteristics are retrieved for letter image similarity characterization to accurately and rapidly describe the many pictorial principles of the manual alphabet's letters [9]. Feature Extraction can be acquired by using PCA (principal component analysis). The unwanted components which does not help to the wanted elements are removed for dimension reduction [10].

D. Classification and Gesture to speech

In this phase, collecting and labelling pictures is the process of classifying them for recognition. The category of the provided input is also predicted by classifiers. Based on their ability to learn, classifiers can be categorized. All the audio recordings of the collected datasets of gestures are stored in the system. when it recognizes the gesture, it gives the output audio.

E. Uml diagrams

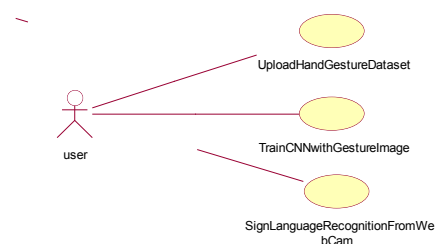


Fig.2. user use case diagram

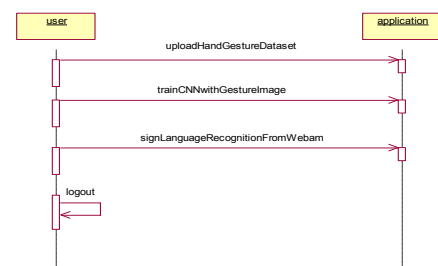


Fig.3. sequence diagram of the proposed method

IV. IMPLEMENTATION

A. Datasets

All the datasets which are collected by the user are stored in the system as shown in Fig.4.



Fig.4. Datasets of different hand gestures

B. Interface

The interface of the proposed work containing upload hand gestures, train CNN with gesture images and Sign Language Recognition from Web Cam are shown in the Fig.5.

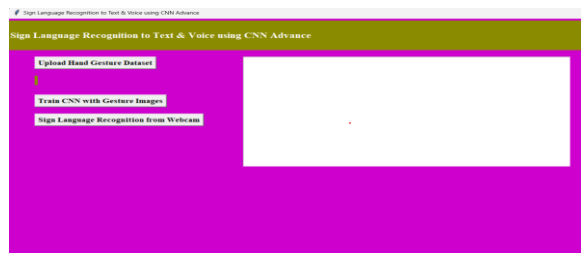


Fig.5. Interface of the proposed system

C. Training CNN

Now click on Upload Hand Gestures and upload your datasets. Then click on Train CNN with gesture images and when it is ready as shown in Fig.6 click on Sign Language Recognition with a webcam to identify the gestures.



Fig.6. Result obtained after training CNN

V. RESULTS

After training the CNN Click on "Sign Language Recognition from Web Cam." Then your webcam opens, and it starts searching for gestures. Make a sign language gesture with your hand now. The CNN starts analysing the hand movements and searches for the perfect match with the provided datasets. It gives the output of the gesture in the form of text and speech as show in the below Fig.7 and Fig.8.

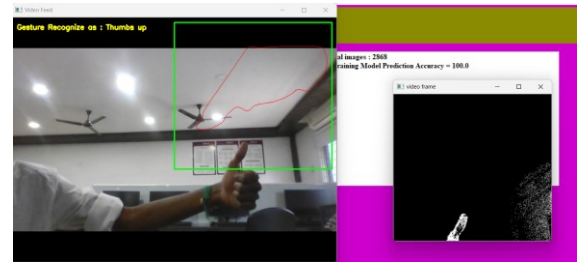


Fig.7. Gesture recognized as Thumbs up

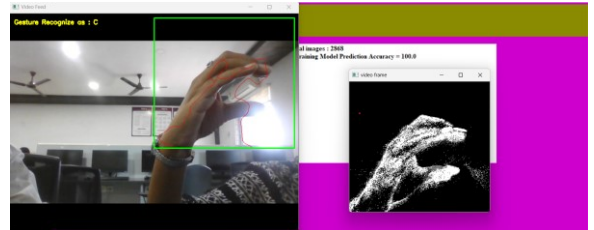


Fig.8. Gesture recognized as C

Both hand gestures are recognised with a perfect accuracy of 100 using CNN algorithm but due to continuous image capturing it takes few seconds to give the accurate result. Accuracy can be calculated by number of successful attempts divided by number of gestures $\times 100$ [11]. Table 1 represents the accuracy acquired by various algorithms.

TABLE 1: ACCURACY COMPARISON

ALGORITHM	ACCURACY
SVM [12]	95.3
KNN [13]	95
PROPOSED CNN	100

VI. CONCLUSION

This proposed work creates a CNN model to acknowledge sign language. By the use of 3D convolutions, the algorithm achieves and retrieves both temporal and spatial features. Convolution and subsampling are carried out individually after the created deep architecture gathers various types of information from nearby input frames. Information from each channel is combined to create the final feature representation. These feature representations are classified using a multilayer perceptron classifier. For the similar datasets, this study compares contrast CNN and GMM-HMM. The outcomes of the experiment shows how successful the suggested approach is.

VII. FUTURE SCOPE

Future studies might concentrate on phrase creation and the identification of dynamic ISL gestures. The system's portability will allow it to be expanded to include smartphones and smart home appliances. In order to attain interaction and usefulness, hand gesture research must outperform present performance in terms of robustness and speed. Extracting characteristics that would recognize each sign regardless of source, color, and lighting conditions requires more attention. For ISL word and sentence level recognition, one can create a model. For this, a system that

can detect changes in temporal space will be required. This can bridge the communication gap for people who are deaf or dumb by developing a comprehensive solution.

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